

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 1 of 10

(1) Research Program/Project/Study Title: EFFECT OF BIO ORGANICS SUPPLEMENTS IN THE PRODUCTIVE PERFORMANCE AND CARCASS YIELD OF DARAG NATIVE CHICKEN

Author's Name:	Dr. Maryneth B. Barrios
Academic Rank:	Associate Professor V
Co-Authors (with academic rank):	Dr. Leah A. Ingalla/ Asso. Prof V
Campus/Satellite College/Address:	Dumara Satellite College/ LRDC
Telephone/Cell Phone/Fax Number:	09070287765
E-mail Address:	mbbarrios@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Dumarao	
Province:		Capiz	
Region:		VI	
(5) Classification		(6) Mode of Implementation ☒ Single Authorship/Proponent	

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	January 26, 2023
			Page	Page 2 of 10

Research: ☒ Basic ☐ Applied	Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization	☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
---	---	---

(7) Sector/Commodity Livestock	(8) Discipline/Field of Specialization Native Chicken
---	--

(9A) Abstract

Philippine native chickens are well known as resistant to diseases, unique taste and flavor and ability to utilize farm by products. However, one of the major constraints in production of this chickens is the smaller slaughter, hence this study was conducted to evaluate the effects of bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) in the productive performance of Darag native chickens and its carcass yield. The study was conducted at CapSU Darag Native Chicken Research Project from October 2023 to April 2024. A total of 75 heads of Darag native chicken ages 4 months in both sexes were used in the study. The study was arranged in Completely Randomized Design and data analysis was done using ANOVA and LSD. Five treatments were employed in this study: Plain water (Control), 10ml fermented Tawa tawa leaves + 20 ml LABS, 15 ml fermented Tawa tawa leaves + 20 ml LABS, 20 ml LABS + fermented Tawa tawa leaves + 20 ml LABS and Commercial LABS. The result of the study showed that supplementation of combined bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) has no significant effect in the productive performance of Darag native chicken. But the use of 15 ml fermented Tawa – tawa + 20 ml LABS can improved the slaughter weight and spleen weight of Darag native chicken.

(9B) Keywords: *Euphorbia hirta*, fermented plant juice, Lactic acid bacteria serum, Tawa-tawa,

(10) Introduction

Darag native chickens are the strain of Philippine native chickens which originally found in Western Visayas. Darag are typically raised in the backyards of rural households. They are commonly grown in small numbers of up to about 24 hens for egg production. They are well known for their adaptability to local agro climatic conditions, hardiness, ability to utilize farm by products, tasty meat and resistance to diseases (PCAARD, 2007). This Darag native chicken gaining its popularity in the Philippines because of its have a unique, and arguably better, taste as compared to the commercial varieties. However, despite of being popular as, one of the problem in raising of it, is the smaller body size at slaughter age. Many researchers had already conducted several studies for different supplements that can be used to enhance the growth of this native chicken. Some of this supplements are the the use of Lactic acid bacteria serum (LABS) and Fermented Plant Juice (FPJ).

LABS(Lactic Acid Bacteria Serum) used as an antibiotic in poultry animals.LAB are safe microorganisms with abilities to produce diffent inhibitory compounds such as bacteriocins,

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 3 of 10

organic acids as lactic acid, hydrogen peroxide, diacetyl, and carbon dioxide. LAB can inhibit harmful microorganisms with their arsenal, or through competitive exclusion mechanism based on competition for binding sites and nutrients. LAB endowed with specific enzymatic functions (amylase, protease) can improve nutrients acquisition as well as animal immune system stimulation (Nuria 2019).

Herbal extracts are being used as feed additives to improve animal performance especially under the intensive management systems. Both health status and general performance can be supported by plant extracts. The positive impact on animals health and performance could be achieved through the stimulation of appetite and feed intake, the improvement of endogenous digestive enzyme secretion, activation of immune response and, antiviral, antioxidant and antihelminthic actions(Omar, 2016). The use of fermented plant juice (FPJ) is one of the process used to utilize this herbal extracts to help animals.

Euphorbia hirta, locally known as tawatawa, has been used in the Philippines as early as 1948 for treating fever. Since the 1980s, it has also been used for sore eyes, cough, diabetes, dengue, malaria, gastritis, and diarrhea. *Euphorbia hirta* is often used traditionally for female disorders, respiratory ailments (cough, coryza, bronchitis, and asthma), worm infestations in children, dysentery, jaundice, pimples, gonorrhea, digestive problems, and tumors. It is reported to contain alkanes, triterpenes, phytosterols, tannins, polyphenols, and flavonoids (Kumar et al., 2010). The study of Hashemi et al., 2012 on broiler chickens showed a positive impact on FCR and their combination displays antioxidant and antibacterial characteristics in broiler chickens. Another study of Zukifli et al., 2010 on the *Euphorbia hirta* and virginiamycin supplementation to the diet on performance of broiler chickens shows also improved over all FCR results. With these, utilization of this tawa tawa or *Euphorbia hirta* was done on Darag native chicken, hence this study was conducted.

Generally, the study was conducted to evaluate the effect of the combined LABS and Tawa tawa fermented plant juice extract on the productive performance and carcass yield of Darag native chickens. Specifically, the study targets to find out the average feed consumption, gain in weight, feed conversion ratio and the net profit of Darag native chicken and to evaluate its carcass yield in terms of slaughter weight, dressed weight, dressing percentage, major and minor cuts, liver, spleen and gizzard weight and the color of the carcass, skin, and abdominal fats.

(11) Materials and Methods or Methodology

Materials

The following materials and equipment were used in the study: 75 heads of 2 months old and both sexes of Darag Native Chicken, feeder and watering troughs, commercial feeds and probiotics, digital weighing scale, lactic acid bacteria serum, and rearing cages, glutinous rice, molasses, plastic bucket, strainer, 10 liters of fresh milk, fermented tawa-tawa plant juice, knives, basin, chopping board, and camera for documentations.

Methods

Experimental Treatments. The study was composed of five treatments which consist of the following: Treatment I- Plain water; Treatment II- 10mL of fermented Tawa- tawa and 20ml formulated LABS in one liter of water; Treatment III-15mL of fermented Tawa-tawa and 20ml formulated LABS in one liter of water; Treatment IV - 20mL of fermented Tawa-tawa and 20ml formulated LABS in one liter of water and Treatment V – Commercial LABS.

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 4 of 10

Experimental Lay-out and Design. The study was conducted in a Completely Randomized Design (CRD). Experimental birds were divided into fifteen groups that correspond to five treatments and three replications as shown in figure below.

Preparation of Housing and Equipment. A total of 15 cages with a floor space area of 1 square meter each were prepared to accommodate five experimental birds to represent one experimental unit. The cages, feeders and waterers were cleaned with soap and water and disinfected using common commercial disinfectant available in the market. The cages and equipment were allowed to air – dry for one week to ensure that the active ingredients were evaporated. The cleanliness of the cages and other materials were also maintained.

	T ₁ R ₁	T ₄ R ₂	T ₄ R ₃	T ₃ R ₃	T ₄ R ₁	T ₂ R ₂	T ₁ R ₃
T ₅ R ₂	T ₂ R ₃	T ₁ R ₂	T ₃ R ₁	T ₂ R ₁	T ₃ R ₂	T ₅ R ₁	T ₅ R ₃

Figure 1.0. Experimental lay-out and distribution of experimental birds in pen arranged in Completely Randomized Design.

Procurement and Quarantine of Experimental Animals. Experimental birds (pure Darag native chickens) were procured from the WVSU Darag Native Chicken Research Project. The birds were quarantined for two weeks in the quarantine house away from the research site. They were also given with multivitamins and electrolytes to recover from stress during transport. After the quarantine period, they were transferred to their assigned pens for data gathering.

Feeds and Feeding. Commercial grower ration (mash or crumble) were given for the whole duration of the study. Experimental birds were fed twice daily; one in the morning and one in the afternoon.

Provision of Water. The experimental birds were provided with fresh and clean tap water every day. Waterers were cleaned and replaced when found dirty. Antibiotic and vitamins were given during the stressful conditions particularly before and after body weights were taken and during blood collection.

Health Management of Experimental Birds. Prior to the start of the study, all experimental birds were given with Newcastle disease vaccine to avoid disease outbreak. However, they were not given with any dewormer in preparation of the study. During stress days such as weighing, blood collection, and in rainy seasons, they were given with anti-stress supplements.

Collection and Preparation of fermented Tawa-tawa (*Euphorbia hirta*) plant juice. The collection and preparation of fermented tawa-tawa plant juice involved several steps. Initially, the fast-growing tips of the plants were collected, preferably before sunrise and avoiding rainy periods. These plant materials were not rinsed to preserve surface microorganisms. Subsequently, the collected material was cut into 2- to 3-inch pieces and weighed. An equal weight of brown sugar was then added and thoroughly mixed with the plant material to facilitate the osmotic process, which draws out the plant juices. The mixture of plant material and brown sugar was packed into a clear glass or food-grade polyethylene (PE) plastic container. The container was covered with a breathable material to allow air exchange while being secured to

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 5 of 10

prevent pests and contaminants from entering. Stored in a well-ventilated area away from light and extreme temperatures, the container was periodically checked after 24 hours to ensure the mixture settled to about two-thirds of its volume, with adjustments made if necessary. Fermentation typically occurred over 3 to 7 days, during which bubbles formed, indicating active fermentation. Completion of fermentation was marked by the plant material floating, a light alcohol smell, and a sweet taste of the liquid. Following fermentation, the plant material was separated from the liquid using a colander or strainer. The resulting liquid, now FPJ, could be used immediately or stored in a loosely covered container to allow gas escape. For long-term storage, an equal amount of brown sugar by weight was added to the FPJ to prevent souring.

Preparation of Lactic acid bacteria serum. Before making Lactic Acid Bacteria Serum (LABS), the first step is to prepare the rice wash that were used. This involves collecting a liter of rice wash and storing it in a clean container that is about two-thirds full, allowing an air gap crucial for the fermentation process. Despite lactobacilli being anaerobic organisms, this air gap facilitates fermentation. The container is tightly covered to prevent contamination, labeled with the preparation date, and stored in a cool, dry place. After 5-7 days, the fermentation process is monitored, and the fermented rice wash is harvested. For the production of LABS, one liter of fermented rice wash is strained into a clean container and mixed with 10 liters of milk. The mixture is tightly covered and marked with the preparation date. After another 5-7 days, the resulting light-yellow liquid is harvested and filtered. The refined LABS is then stored in a refrigerator at 1-15°C, where the bacteria enter a dormant state, remaining viable but not actively consuming nutrients or proliferating. For room temperature storage, LABS is mixed with molasses in equal proportions (1:1 ratio) in a 3- 5-liter container, ensuring the bacilli have food during storage, thus maintaining their viability.

Administration of treatments. Experimental birds were weighed a day before the start of the study. Administration of treatments was conducted daily. Each morning, the birds were provided with the given treatments mixed with 10 ml of water per cage. This consistent daily administration ensured that all chickens received the appropriate dosage necessary to evaluate the impact of these treatments on their carcass quality.

Data Gathering Procedures

Productive Performance. Productive performance were determined to measure the effect of combined bio organics (LABS and FPJ) by determining the gain in weight, feed consumption, feed conversion ratio, and the net profit of birds (Ozaraga, 2012). Below are the procedures in data gathering of productive performance of experimental animals:

Average Feed Consumption (AFC). The feed consumption of the birds were determined by subtracting the weight of the left-over feeds from weight of feeds given. This was done from the start of experimental feeding until the day of termination of the study per replicate basis. This was computed using the formula:

$$AFC = \frac{\text{Amount of feeds given} - \text{Weight of left-over feeds}}{\text{Weight of birds}} \times 100$$

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 6 of 10

Number of birds per replicate

Gain in Weight. The gain in weight of the birds (grams) were determined by weighing them at the start of the feeding period. This was taken by subtracting the final weight to the initial weight of the birds.

Feed Conversion Ratio. Feed conversion ratio were obtained by dividing the average feed consumption by the average gain in weight. This was computed using this formula.

$$\text{FCR} = \frac{\text{Average Feed Consumption}}{\text{Average Gain in Weight}}$$

Net Income. To get the net income of the native chicken as affected by the different levels of lactic acid bacteria serum, the data were calculated by subtracting the total expenses from the total sales obtained from the live chickens.

Carcass Yield. The data gathered in this parameter were done after the three months of bio organics supplementation to Darag native chickens. Data gathered were the following:

1. **Slaughter weight (grams).** The weight of the bird before slaughter time. This were obtained by weighing the Darag native chicken before it is slaughtered.
2. **Dressed weight (grams).** This were taken after plucking the feathers, and removing the heads, feet entrails of the chickens.
3. **Dressing percentage (%).** This were obtained by dividing the dressing weight by the slaughter weight of the chickens multiplied by 100%.
4. **Major cuts weight (grams).** This were taken by weighing the weight of back, breast, wings, and legs.
5. **Minor cuts weight (grams).** This were taken by weighing the weight of head, feet and neck.
6. **Liver, Gizzard and spleen weight (grams).** This were taken by weighing the weight of head, feet and neck.
7. **Color.** This were obtained by getting the color of the carcass, skin and abdominal fats.

Data Analysis Procedures

Data collected were analyzed using the Analysis of Variance (ANOVA). Differences in treatment means were compared using the Least Significant Difference (LSD). The result were interpreted at five percent level of significance.

(12) Results and Discussions

Effect of Bio organics (LABS and Fermented Tawa – tawa FTT) Supplementation in the Productive Performance of Darag Native Chickens

Table 1.0 presents the productive performance of Darag native chicken supplemented with combined Bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) at different levels. The result of the study showed that the feed consumption of Darag native chicken range from 4409.00 grams to 4421.00 grams, gain in weight ranged from 503.33 grams to 646.00 grams, FCR of 7.26 to 9.51 and a net income of 2.41 pesos to 156.79 pesos. Result of the study further shows that supplementation of biorganics to Darag native chicken diets

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 7 of 10

has no significant effect ($P>0.05$) in the productive performance of the chickens. This results conforms to the study of Liu et al., 2018 to the effect of lactic acid bacteria on the growth performance of broiler chicken. Their study showed that inclusion of LAB to broilers diet can increased body weight gain and feed intake of chickens but has a similar effect to the control group.

Table 1.0 Average productive performance of Darag native chickens supplemented with combined Bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) at different levels.

Treatments	Productive Performance			
	Feed Consumption (g) ^{ns}	Gain in Weight (g) ^{ns}	Feed Conversion Ratio ^{ns}	Net Income
Plain Water	4415.00	646.00	7.26	156.79
10ml FTT and 20 ml LABS	4409.00	542.33	8.17	97.86
15ml FTT and 20 ml LABS	4415.33	503.33	9.00	97.05
20ml FTT and 20 ml LABS	4421.00	519.33	8.66	92.14
Commercial LABS	4416.67	506.33	9.51	2.41

ns-not significant

Effect of Bio organics (LABS and Fermented Tawa – tawa FTT) Supplementation in the Carcass Yield of Darag Native Chickens

Table 2.0 and 3. 0 presents the the carcass yield of Darag native chicken as affected by the supplementation of bio organics. Data showed that chickens given with 15 ml fermented tawa – tawa and 20 ml LABS obtained the highest slaughter weight and spleen weight and was significantle different from other treatments ($P>0.05$). On the other hand, the slaughter weight of Darag native chicken supplemented with combined bio organics ranged from 1,333.33 grams to 1,600.00 grams; the dressed weight ranged from 1,026.67 grams to 1, 383.33 grams; dressing percentage of 71.57% to 90.92%; major cuts weight of 229.50 grams to 266.33 grams; minor cuts of 37.22 grams to 76.00 grams; spleen weight of 5.00 grams to 6.00 grams; liver weight of 23.33 grams to 26.67 grams and gizzard weight of 28.33 grams to 31.67 grams. Moreover, the color of the skin and carcass of the birds were all white while abdominal fats has white and light yellow color.

Table 2.0 Carcass yield of Darag native chickens supplemented with combined Bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) at different levels.

Treatments	Slaughter Weight*	Dressed Weight ^{ns}	Dressing Percentage ^{ns}	Major Cuts Weight ^{ns}	Minor Cuts ^{ns}	Splee n*	Liver ^{ns}	Gizzard ^{ns}
Plain water	1433.33 ^d	1140.00	79.52	263.75	37.22	5.00 ^b	23.33	28.33
10mL of FTT and 20ml FLABS	1333.33 ^c	1166.67	86.90	264.25	71.11	5.00 ^b	23.33	31.67
15mL of FTT and 20ml FLABS	1600.00 ^a	1170.00	73.13	264.58	70.00	6.00 ^a	25.00	31.67
20mL of FTT and 20ml FLABS	1433.30 ^d	1026.67	71.57	229.50	76.00	5.33 ^b	26.67	31.67
Commercial LABS	1533.33 ^b	1383.33	90.92	266.33	65.52	5.00 ^b	25.00	31.67

ns – not significant * significant at 5% Means of the same letter are comparable with each other

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 8 of 10

Table 3.0 Color of skin, carcass and abdominal fats of Darag native chickens supplemented with combined Bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) at different levels.

Treatments	Skin Color	Carcass Color	Abdominal fats Color
Plain water	White	White	White
10mL of FTT and 20ml FLABS	White	White	White
15mL of FTT and 20ml FLABS	White	White	White, Light yellow
20mL of FTT and 20ml FLABS	White	White	White, Light yellow
Commercial LABS	White	White	White, Light yellow

The results of this study conforms to the result of the study of Sjofjan et al., 2021 that supplementation of probiotics in broiler diet increased ($p < 0.001$) the weight of liver, spleen, gizzard, bursa of fabricius and carcass yield of broilers.

(13) Conclusions and Recommendations

Based on the results of the study, the researcher concludes that supplementation of combined bio organics (Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract) has no significant effects in the productive performance of Darag native chickens. But the use of 15 ml and 20 ml per 1 liter of water of Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract as supplements to Darag native chickens can increased the slaughter weight and the spleen weight of chickens.

Based on the findings of the study, the researchers recommends the use of 15 ml and 20 ml per 1 liter of water of Lactic acid bacteria serum and fermented *Euphorbia hirta* juice extract as supplements to Darag native chickens to increased the slaughter weight and the spleen weight of chickens. Explore another study of using different plants as fermented juice as supplements of animals.

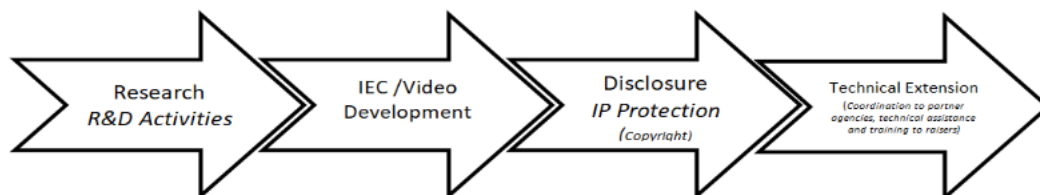
(14) Output

Expected Output	Details/ Activities Conducted	Target
People Services	Train/ orient native chicken growers on the use of fermented plant juice as supplements to native chickens.	30 raisers
Products	Production of fermented plant juice supplements for native chickens	10 liters
Places and Partnerships	Forge linkages and partnerships to LGU's and other SUC's for future collaboration of the further study of serpentina plant	2
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials for preparation and utilization of Serpentina plant.	2
Patents/IP	File 2 copyrights for the developed IEC materials.	2

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 9 of 10

Policy	Develop one policy inputs to for the utilization of fermented plant juice as feed supplements.	1	
Impact	Help native chicken raisers to improved native chicken production		

(15) Technology Transfer Activity Conducted



(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments

The researcher would like to thank the CapSU RDE and LRDC for funding the study.

(18) References

- Abella, L.** (2013). Efficacy of Ethnobotanical Extracts as Anthelmintics against Intestinal Parasites of Darag Native Chicken (*Gallus gallus domesticus*). Unpublished Master's Thesis. Capiz State University, Mambusao, Capiz.
- Hashemi,S.R., I. Zulkifli, H. Davoodi, Z. Zunita, M. Ebrahimi.** (2012). Growth performance, intestinal microflora, plasma fatty acid profile in broiler chickens fed herbal plant (*Euphorbia hirta*) and mix of acidifiers, *Animal Feed Science and Technology*, Volume 178, Issues 3–4. <https://doi.org/10.1016/j.anifeedsci.2012.09.006>.
- Kumar, S., Malhotra, R., & Kumar, D.** (2010). *Euphorbia hirta*: Its chemistry, traditional and medicinal uses, and pharmacological activities. *Pharmacognosy reviews*, 4(7), 58–61. <https://doi.org/10.4103/0973-7847.65327>
- Liu, N., Ding, K., Wang, J., Gu, K., and Wang, J.** (2018). Effects of lactic acid bacteria and smectite after aflatoxin B₁ challenge on the growth performance, nutrient digestibility and blood parameters of broilers. *Journal of Animal Nutrition and Physiology*. <https://onlinelibrary.wiley.com/doi/abs/10.1111/jpn.12901>

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	January 26, 2023
			Page	Page 10 of 10

PCARRD. (2007). Profitability Analysis : 200 Hen Module of Native Chicken Production. 1st Edition. ISSN: 1908 – 8043. DOST, Los Baños, Laguna.

Sjofjan, O., Adli, D. N., Harahap, R. P., Jayanegara, A., Utama, D. T., & Seruni, A. P. (2021). The effects of lactic acid bacteria and yeasts as probiotics on the growth performance, relative organ weight, blood parameters, and immune responses of broiler: A meta-analysis. *F1000Research*, 10, 183. <https://doi.org/10.12688/f1000research.51219>.

Ozaraga, M. S. (2012). Efficacy of Ethnobotanicals as Anthelmintics in the Productive Performance of Darag Native Chicken. Unpublished Dissertation. Capiz State University – Burias, Mambusao, Capiz.